



Understanding constructing a circle

Task A: Properties of a circle

1) You will need a blank piece of A4 paper for question 1. It will not fit on this sheet.

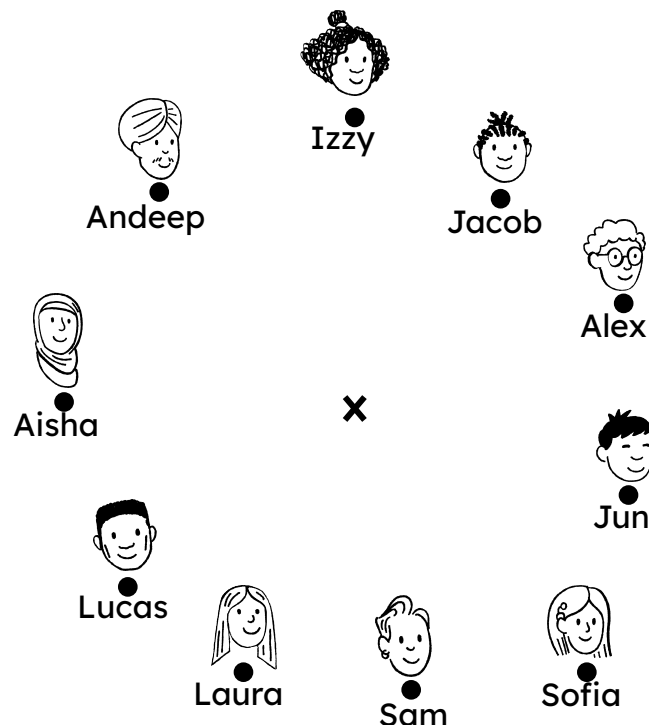
Mark a cross in the middle of your plain piece of paper.

a) Mark 8 points that are exactly 7 cm away from the cross.

b) Draw a circle with radius 7 cm and with the centre at the cross.

c) What do you notice about the points from part (a) and the circle from part (b)?

2) Aisha marks a cross on a piece of paper and marks a point exactly 5 units away from it. She asks everyone else to mark a point 5 units away from the cross.

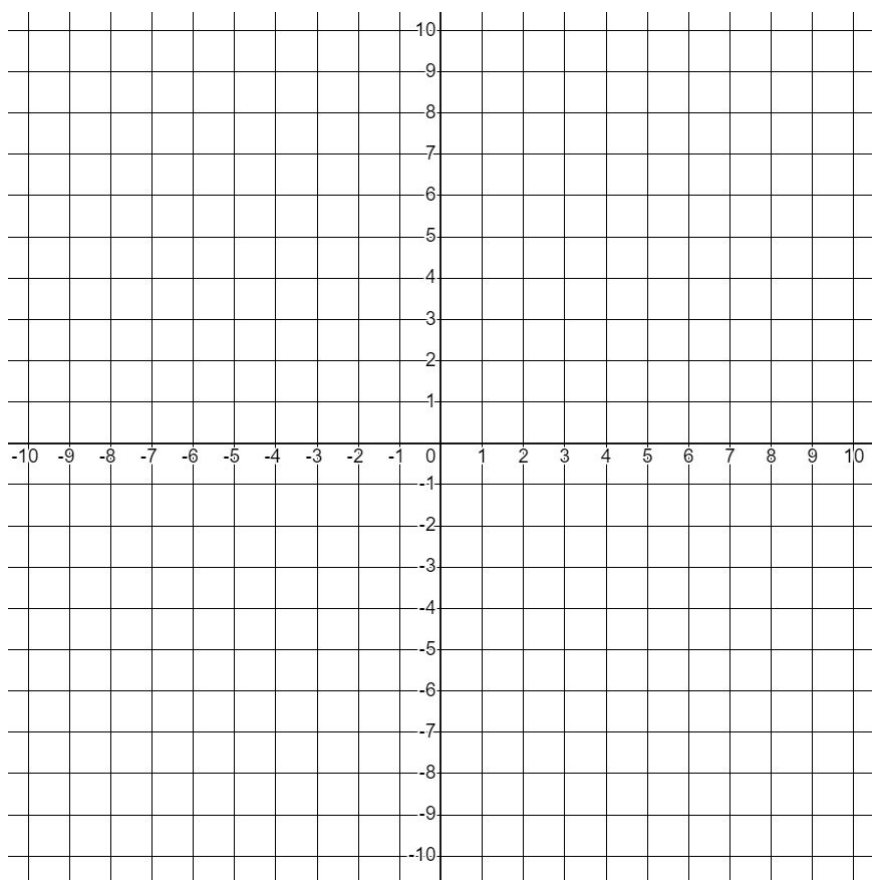


a) How could you use a pair of compasses to check whether everyone's point is the correct distance away from the cross?

b) Whose point is too close to the cross?

c) Whose point is too far from the cross?

3) Find the coordinates of 12 points which are exactly 10 units from the origin.



Task B: Properties of an arc of a circle

1) W, X, Y and Z are arcs from four circles.

A, B, C and D are four possible centres for these circles.

Match the arcs to their centres.

